

AmExpert 2019 - ML Hackathon

Problem Statement

The organizer is a credit card company which has sourced data from one of its merchant partners on past marketing campaigns. The customers are typically offered a coupon for discounted purchases in a campaign. Data has been provided for last 18 such campaigns. It includes information on contacts, transactions, demographics, coupon and item details. The target variable is redemption status on any such marketing contact defined at campaign-customer-coupon level. The predictions have to be made for 10 such campaigns into the future.

Approach

At a high level, the solution involves building a few GBM models on derived features, that have been created by looking at interactions of various customer attributes, coupon attributes etc. The next sections outlines this approach in detail, in particular how these features were created and how the data was subsequently modeled.

Data Preprocessing:

The data files were very mostly clean to begin with and required minimal pre-processing.

- **campaign_data:** The start and end_dates were in dd/mm/yy format which were sometimes parsed incorrectly by pandas. This was handled by converting the data yyyy-mm-dd format, a format that was being used in transaction data as well.
- **item_data:** brand, brand_type and category were label encoded
- **customer_demographics:** This dataset had missing values. Additionally, age_range, marital_status, no_of_children and family_size had to be encoded.

Feature Engineering:

This section outlines how the features were created by looking at the data from various views.

1. **campaign:** campaign_type, day, month and duration features

2. **count**: Count-based features on customer, campaign, coupon and their combinations as they appear in train and test data. The intuition is to capture the volume and diversity of each campaign.
3. **coupon_spend**: Merged coupon_item_mapping to transaction and summarized price, quantity, other_discount and coupon_discount. The idea is gauge the popularity of any coupon. A coupon is essentially composed of various items. By looking at the transaction data in this fashion, we can pick on signals of how popular the coupon can be with potential customers.
4. **coupon**: Merged coupon_item_mapping and coupon_data. Aggregated information the volumes of items, brands and categories present in each coupon. Idea is to capture trends based on coupon construct.
5. **customer**: encoded the categorical features in demographics data. Also, summarized the spend behavior by looking at transaction data and summarizing price, quantity, item information for each customer. Idea is to create a customer spend profile.
6. **similarity**: It computed Jaccard distance between the item buckets of customer purchases and coupon construct. The hypothesis is - if there is a large overlap between coupon and customer's typical cart, then the chances of redemption may go up.
7. **tranx-item**: Removed the rows that involved coupon discount from transaction data to prevent leakage. The calculated customer-coupon affinity by aggregating the customer-item information across each coupon. The features included # purchases, total amount, maximum and price etc.
8. **tranx-brand**: Same as above but operates at brand level
9. **tranx-category**: Same as above but operates at category level
10. **tranx-time**: For any campaign contacts, summarized the transactions which occurred before the campaign start date. This can help us leverage the coupon discounts information in a lagged fashion.

Model

As stated above, the final solution relies on three GBM models. The details are as follows:

1. **Training**: The dataset is split based on campaign_id. campaign_id = 13 is used for validation as its response rate is closest to the overall data.
2. **Scoring**: After the model is tuned using the above train-valid schema, it's trained again, using the same hyper-parameters, on the full dataset (train+valid). This keeps the information loss at minimum.
3. **Model-1**: A lightgbm model that drops all the id related features and relies only in derived features.
4. **Model-2**: A lightgbm model that uses customer_id as a categorical feature along with derived features. There is a huge overlap between train and test customer set. Hence,

customer_id can be used as a feature.

5. **Model-3:** Same as model 1 but uses catboost. It brings some diversity to the blend (discussed below).

6. **Blend:** Simple rank average of the above three model scores.

Closing Thoughts

The solution relies on capturing the relevance of a coupon to any particular customer by leveraging personalization using the features created from transaction data. Further improvements can be realized by exploring more sophisticated feature engineering approaches in the same direction using cosine similarity, matrix factorization, graph based solutions etc.